

- 3.12 **Recall that an isotope above the curve has too many neutrons to be stable and will undergo  $\beta^-$  decay**
- 3.13 **Recall that an isotope below the curve has too many protons to be stable and will undergo  $\beta^+$  decay**
- 3.14 **Recall that the proton and neutron each contain three particles called quarks**
- 3.15 **Describe the arrangement of up and down quarks in protons and neutrons**
- 3.16 **Use given data to explain the arrangement of up and down quarks in protons and neutrons in terms of charge and mass**
- 3.17 **Explain  $\beta^-$  decay as a process that involves a down quark changing into an up quark (a neutron becomes a proton and an electron)**
- 3.18 **Explain  $\beta^+$  decay as a process that involves an up quark changing into a down quark (a proton becomes a neutron and a positron)**
- 3.19 Recall that nuclei that have undergone radioactive decay often undergo nuclear rearrangement with a loss of energy as gamma radiation
- 3.20 Describe the dangers of ionising radiation in terms of tissue damage and possible mutations
- 3.21 Explain the precautions taken to ensure the safety of people exposed to radiation, including limiting the dose for patients and the risks to medical personnel
- 3.22 Compare and contrast the treatment of tumours using radiation applied internally or externally
- 3.23 Describe palliative care including the use of radiation in some instances
- 3.24 Explain some of the uses of radioactive substances in diagnosis of medical conditions, including PET scanners and tracers
- 3.25 Explain why isotopes used in PET scanners have to be produced nearby

## Topic 4

### Motion of particles

- 4.1 Discuss how instruments, including particle accelerators, can help scientists develop better explanations about the physical world
- 4.2 Discuss reasons for collaborative, international research into big scientific questions, including particle physics
- 4.3 Explain how for motion in a circle there must be a resultant force known as a centripetal force that acts towards the centre of the circle
- 4.4 Explain how particle accelerators called cyclotrons cause charged particles to move in a circular or spiral path, due to a magnetic field